

Fig. 3: An actual-size PCB layout for the laser based security lock circuit

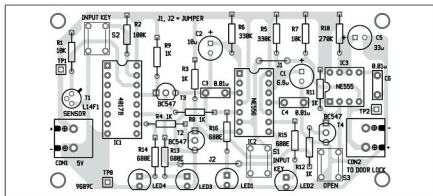


Fig. 4: Component layout of the PCB

timer of IC2. Its output pin 9 goes high for two seconds. The high state is indicated by the glowing of LED2. When LED2 goes off, you press switch S1 and release it. The first timer is triggered again and its output is high for four seconds. During this time, you send another five pulses of laser beam towards the sensor. At the fifth pulse, LED3 glows and transistor T4 conducts. This action triggers IC3 and its output pin 3 goes high for ten seconds. That is, the lock opens for ten seconds.

The lock driver circuit is connected to output pin 3 of IC3. The circuit is powered off a 5V DC supply. Nor-

mally the electric lock or electric strike works off a 12V DC. To make the circuit simple, the lock driver section is not shown here.

Construction and testing

An actual-size, single-side PCB layout of the laser-based security lock is shown in Fig. 3 and its component layout in Fig. 4.

Mount the components on the PCB to avoid any assembly errors. Enclose the circuit in a suitable box. Mount switches S1, S2 and S3 at appropriate locations. If the circuit is to be used as a door lock, photo sensor T1, switches S1

PARTS LIST

Semiconductors:

- IC1 - 4017B decade counter
- IC2 - NE555 dual timer
- IC3 - NE555 timer
- T1 - LI4F1 photo Darlington transistor sensor

T2-T4

- BC547 npn transistor

LED1-LED4

- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R7 - 10-kilo-ohm
- R2 - 100-kilo-ohm
- R3-R4, R8, R9, R11, R12 - 1-kilo-ohm
- R5, R6 - 330-kilo-ohm
- R10 - 270-kilo-ohm
- R13-R16 - 680-ohm

Capacitors:

- C1 - 6.8 μ F, 16V electrolytic
- C2 - 10 μ F, 16V electrolytic
- C3, C4, C6 - 0.01 μ F ceramic disk
- C5 - 33 μ F, 16V electrolytic

Miscellaneous:

- CON1, CON2 - 2-pin terminal connector
- S1, S2, S3 - Tactile switch
- 12V electric strike or lock

and S2 should be mounted outside the door frame. The photo sensor should be enclosed in the box with a suitable contraption so that it can receive the laser beam properly. Switch S3 should be installed on other side of the wall near the door frame.

While pressing switches S1 and S2 simultaneously, point the laser torch toward the photo transistor sensor T1 and press on/off button of the laser torch five times within four seconds. Release switch S1 while pressing switch S2. Wait for two seconds till LED2 goes off. Keeping S2 pressed, press S1 and release it, press on/off button of laser torch five times again. The lock will open for ten seconds and then close automatically. The lock can also be opened from the inside (for ten seconds) by pressing switch S3.

For troubleshooting, refer the test points table. ●



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